

Longitudinal Data Analysis with R Statistical Software

Day 3: Mixed-effects models

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Vanderbilt-Nigeria Biostatistics Workshop

April 27 – May 1, 2026

Day 2 review

Mixed-effects models

- ★ Contrast outcomes both within and between individuals
 - Assume that each subject has a regression model characterized by subject-specific parameters; a combination of
 - ▶ **Fixed-effects** parameters common to all individuals in the population
 - ▶ **Random-effects** parameters unique to each individual subject
 - Although covariates allow for differences across subjects, typically cannot measure all factors that give rise to subject-specific variation
 - Subject-specific random effects induce a correlation structure

(Laird and Ware, 1982)

Set-up

For subject i the mixed-effects model is characterized by

$$y_i = \{y_{i1}, y_{i2}, \dots, y_{im_i}\}^T \quad \text{Outcomes}$$

$$\beta = \{\beta_0, \beta_1, \beta_2, \dots, \beta_p\}^T \quad \text{Fixed-effects parameters}$$

$$x_{ij} = \{1, x_{ij1}, x_{ij2}, \dots, x_{ijp}\} \quad \text{Covariates}$$

$$X_i = \{x_{i1}, x_{i2}, \dots, x_{im_i}\}^T \quad \text{Design matrix for fixed effects}$$

$$b_i = \{b_{i0}, b_{i1}, b_{i2}, \dots, b_{iq}\}^T \quad \text{Random-effects parameters}$$

$$z_{ij} = \{1, z_{ij1}, z_{ij2}, \dots, z_{ijq}\} \quad \text{Covariates}$$

$$Z_i = \{z_{i1}, z_{i2}, \dots, z_{im_i}\}^T \quad \text{Design matrix for random effects}$$

for $i = 1, \dots, n$; $j = 1, \dots, m_i$; and $q \leq p$

Linear mixed-effects model

Consider a linear mixed-effects model for a continuous outcome y_{ij}

1. Model for response given random effects

$$y_{ij} = x_{ij}\beta + z_{ij}b_i + \epsilon_{ij}$$

with

- ▶ x_{ij} : vector a covariates
- ▶ β : vector of fixed-effects parameters
- ▶ z_{ij} : subset of x_{ij}
- ▶ b_i : vector of random-effects parameters
- ▶ ϵ_{ij} : observation-specific measurement error

2. Model for random effects

$$b_i \sim N(0, D)$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

with b_i and ϵ_{ij} assumed to be independent

Choices for random effects

Consider the linear mixed-effects models that include

- **Random intercepts**

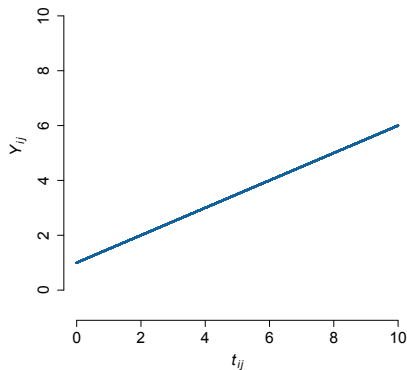
$$\begin{aligned}y_{ij} &= \beta_0 + \beta_1 t_{ij} + b_{i0} + \epsilon_{ij} \\ &= (\beta_0 + b_{i0}) + \beta_1 t_{ij} + \epsilon_{ij}\end{aligned}$$

- **Random intercepts and slopes**

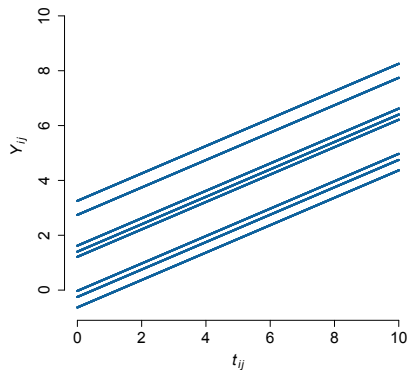
$$\begin{aligned}y_{ij} &= \beta_0 + \beta_1 t_{ij} + b_{i0} + b_{i1} t_{ij} + \epsilon_{ij} \\ &= (\beta_0 + b_{i0}) + (\beta_1 + b_{i1}) t_{ij} + \epsilon_{ij}\end{aligned}$$

Choices for random effects

Fixed intercept, fixed slope

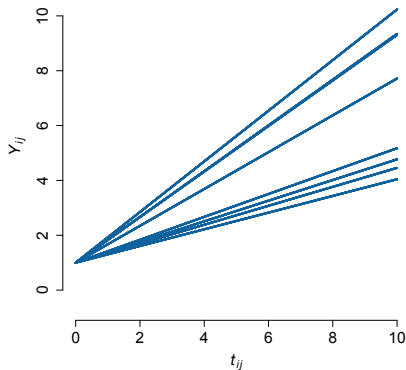


Random intercept, fixed slope

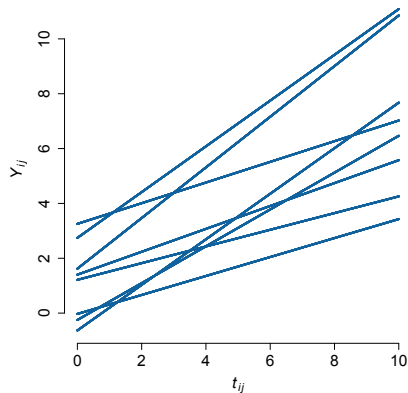


Choices for random effects

Fixed intercept, random slope



Random intercept, random slope



Choices for random effects: D

D quantifies random variation in trajectories across subjects

$$D = \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix}$$

- $\sqrt{D_{11}}$ is the typical deviation in the **level** of the response
- $\sqrt{D_{22}}$ is the typical deviation in the **change** in the response
- D_{12} is the covariance between subject-specific intercepts and slopes
 - ▶ $D_{12} = 0$ indicates subject-specific intercepts and slopes are uncorrelated
 - ▶ $D_{12} > 0$ indicates subjects with high level have high rate of change
 - ▶ $D_{12} < 0$ indicates subjects with high level have low rate of change

$$(D_{12} = D_{21})$$

Induced correlation structure

What is the correlation between measurements on the same subject?

- **Random intercepts model**

- ▶ Assuming $\text{Var}[\epsilon_{ij}] = \sigma^2$ and $\text{Cov}[\epsilon_{ij}, \epsilon_{ij'}] = 0$

$$y_{ij} = \beta_0 + \beta_1 t_{ij} + b_{i0} + \epsilon_{ij}$$

$$y_{ij'} = \beta_0 + \beta_1 t_{ij'} + b_{i0} + \epsilon_{ij'}$$

$$\begin{aligned}\text{Var}[Y_{ij}] &= \text{Var}_b[\text{E}_Y(Y_{ij} \mid b_{i0})] + \text{E}_b[\text{Var}_Y(Y_{ij} \mid b_{i0})] \\ &= D_{11} + \sigma^2\end{aligned}$$

$$\begin{aligned}\text{Cov}[Y_{ij}, Y_{ij'}] &= \text{Cov}_b[\text{E}_Y(Y_{ij} \mid b_{i0}), \text{E}_Y(Y_{ij'} \mid b_{i0})] \\ &\quad + \text{E}_b[\text{Cov}_Y(Y_{ij}, Y_{ij'} \mid b_{i0})] \\ &= D_{11}\end{aligned}$$

Induced correlation structure

- **Random intercepts model** (continued)

$$\begin{aligned}\text{Corr}[Y_{ij}, Y_{ij'}] &= \frac{D_{11}}{\sqrt{D_{11} + \sigma^2} \sqrt{D_{11} + \sigma^2}} \\ &= \frac{D_{11}}{D_{11} + \sigma^2} \\ &= \frac{\text{'Between'}}{\text{'Between'} + \text{'Within'}} \\ &\geq 0 \text{ (and } \leq 1\text{)}\end{aligned}$$

- ▶ Any two measurements on the same subject have the same correlation; does not depend on time nor the distance between measurements
- ▶ Longitudinal correlation is constrained to be positive ($D_{11} \geq 0$, $\sigma^2 \geq 0$)

Induced correlation structure

- **Random intercepts and slopes model**

- ▶ Assuming $\text{Var}[\epsilon_{ij}] = \sigma^2$ and $\text{Cov}[\epsilon_{ij}, \epsilon_{ij'}] = 0$

$$\begin{aligned}y_{ij} &= (\beta_0 + \beta_1 t_{ij}) + (b_{i0} + b_{i1} t_{ij}) + \epsilon_{ij} \\y_{ij'} &= (\beta_0 + \beta_1 t_{ij'}) + (b_{i0} + b_{i1} t_{ij'}) + \epsilon_{ij'}\end{aligned}$$

$$\begin{aligned}\text{Var}[Y_{ij}] &= \text{Var}_b[\text{E}_Y(Y_{ij} \mid b_i)] + \text{E}_b[\text{Var}_Y(Y_{ij} \mid b_i)] \\&= D_{11} + 2D_{12}t_{ij} + D_{22}t_{ij}^2 + \sigma^2\end{aligned}$$

$$\begin{aligned}\text{Cov}[Y_{ij}, Y_{ij'}] &= \text{Cov}_b[\text{E}_Y(Y_{ij} \mid b_i), \text{E}_Y(Y_{ij'} \mid b_i)] \\&\quad + \text{E}_b[\text{Cov}_Y(Y_{ij}, Y_{ij'} \mid b_i)] \\&= D_{11} + D_{12}(t_{ij} + t_{ij'}) + D_{22}t_{ij}t_{ij'}\end{aligned}$$

Induced correlation structure

- **Random intercepts and slopes model** (continued)

$$\begin{aligned} & \text{Corr}[Y_{ij}, Y_{ij'}] \\ &= \frac{D_{11} + D_{12}(t_{ij} + t_{ij'}) + D_{22}t_{ij}t_{ij'}}{\sqrt{D_{11} + 2D_{12}t_{ij} + D_{22}t_{ij}^2 + \sigma^2} \sqrt{D_{11} + 2D_{12}t_{ij'} + D_{22}t_{ij'}^2 + \sigma^2}} \end{aligned}$$

- ▶ Any two measurements on the same subject may not have the same correlation; depends on the specific observation times

Likelihood-based estimation of β

Requires specification of a complete probability distribution for the data

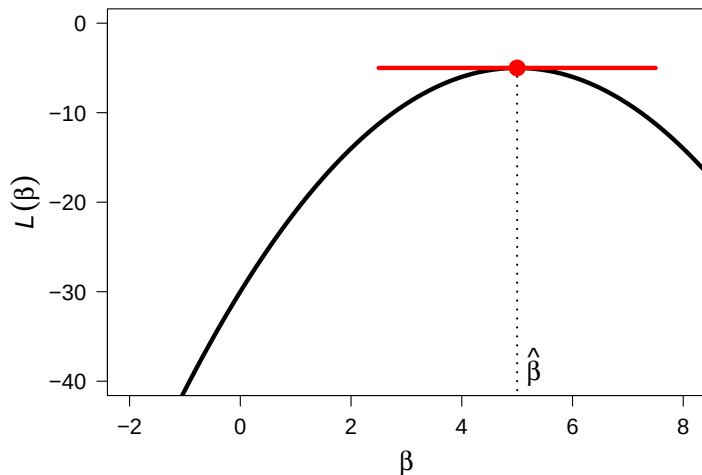
- Likelihood-based methods are designed for fixed effects, so integrate over the assumed distribution for the random effects

$$\mathcal{L}(\beta, \sigma, D) = \prod_{i=1}^n \int f_Y(y_i \mid b_i, \beta, \sigma) \times f_b(b_i \mid D) db_i$$

where f_b is typically the density function of a Normal random variable

- ▶ For linear models the required integration is straightforward because y_i and b_i are both normally distributed (easy to program)
- ▶ For GLMM the integration is difficult and requires either approximation or numerical techniques (hard to program)
- ▶ Conditional likelihood methods treat the random effects as fixed and condition on statistics for them

Likelihood-based estimation of β



Likelihood-based inference for β

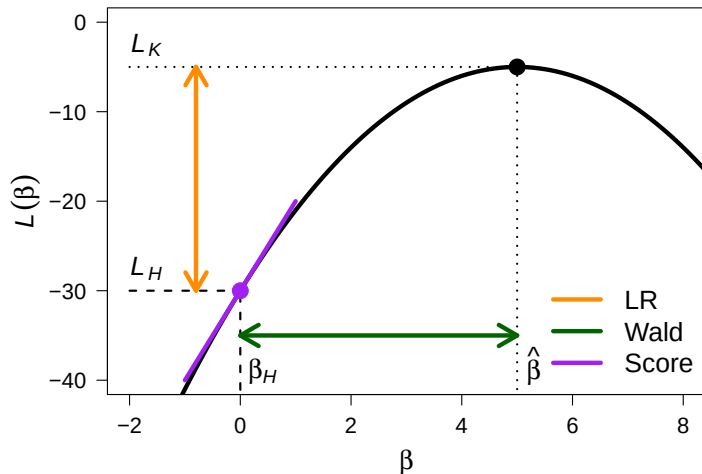
Consider testing fixed effects in nested linear mixed-effects models

$$H: \beta = \begin{bmatrix} \beta_1 \\ 0 \end{bmatrix} \quad \text{versus} \quad K: \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix},$$

i.e., $H: \beta_2 = 0$

- Likelihood ratio test is valid with maximum likelihood estimation
 - ▶ Requires computation under the null and alternative hypotheses
- Likelihood ratio test may not be valid with other estimation methods (REML)
- Wald test (based on coefficient and standard error) is generally valid

Likelihood-based inference for β



Likelihood-based inference for D

Consider testing whether a random intercept model is adequate

$$H: D = \begin{bmatrix} D_{11} & 0 \\ 0 & 0 \end{bmatrix} \quad \text{versus} \quad K: D = \begin{bmatrix} D_{11} & \\ D_{12} & D_{22} \end{bmatrix},$$

i.e., $H: D_{12} = D_{22} = 0$

- Adequate covariance modeling is useful for the interpretation of the random variation in the data
- Over-parameterization of the covariance structure leads to inefficient estimation of fixed-effects parameters β
- Covariance model choice determines the standard error estimates for $\hat{\beta}$; correct model is required for correct standard error estimates
- Generally recommend against this inferential procedure
 - ▶ Specification for the covariance structure should be guided by *a priori* scientific knowledge and exploratory data analysis

Restricted maximum likelihood (REML)

- Standard ML estimation for variance components (σ , D) does not account for estimation of β , which can lead to biased estimates for variance components in small samples (–)
- REML: Estimates variance components after accounting for estimation of β
 - ▶ Corrects for small-sample bias (+)
 - ▶ Likelihood ratio tests for D can be used (+)
 - ▶ Likelihood ratio tests for β cannot be used (–)

Assumptions

Valid inference from a linear mixed-effects model relies on

- **Mean model:** As with any regression model for an average outcome, need to correctly specify the functional form of $x_{ij}\beta$ (here also $z_{ij}b_i$)
 - ▶ Included important covariates in the model
 - ▶ Correctly specified any transformations or interactions
- **Covariance model:** Correct covariance model (random-effects specification) is required for correct standard error estimates for $\hat{\beta}$
- **Normality:** Normality of ϵ_{ij} and b_i is required for normal likelihood function to be the correct likelihood function for y_{ij}
- n sufficiently large for **asymptotic inference** to be valid

★ These assumptions must be verified to evaluate any fitted model

GLMM

	LME	GLMM
Outcome	Continuous	Count, binary
Distribution	Normal	Poisson, binomial
Link	Identity	log, log-odds
Random effects	Intercepts, slopes	Intercepts, slopes
Interpretation	Straightforward	More complicated
Estimation	ML/REML	Numerical approximation
Computation	Fast, stable	Slow, convergence issues
lme4 command	lmer	glmer

Summary

- Mixed-effects models assume that each subject has a regression model characterized by subject-specific parameters; a combination of
 - ▶ Fixed-effects parameters common to all individuals in the population
 - ▶ Random-effects parameters unique to each individual subject
- Estimation and inference can focus both on average outcome levels and trends, and on heterogeneity across subjects in levels and trends
- Subject-specific random effects induce a correlation structure
- Parametric likelihood approach permits use of likelihood ratio test, but requires several assumptions that must be verified in practice

Issues

- Interpretation depends on outcomes and random-effects specification
- GLMM requires that any missing data are missing at random
- Issues arise with time-dependent exposures and covariance weighting

Key points

- Conditional mean regression model
- Model for population heterogeneity
- Subject-specific random effects induce a correlation structure
- Multiple sources of positive correlation
- Fully parametric model based on assumed distributions
- Estimates obtained from likelihood function
- Conditional (fixed effects) and maximum (random effects) likelihood
- Approximation or numerical integration to integrate out b_i
- Requires correct parametric model specification
- Testing with likelihood ratio and Wald tests
- Conditional or subject-specific inference
- Induced marginal mean structure and 'attenuation'
- Missing at random (MAR)
- Time-dependent covariates and endogeneity
- R package lme4

Big picture

- Provide valid estimates and standard errors for regression parameters only under stringent model assumptions that must be verified (–)
- Provide population-averaged or subject-specific inference depending on the outcome distribution and specified random effects (+/–)
- Accommodate multiple sources of correlation (+/–)
- Require that any missing data are missing at random (–/+)